

Large Language Models for Clinical Diagnosis and Treatment Recommendation

Bachelor Semester Project S2 (Academic Year 2025/26), University of Luxembourg

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Abstract

Evaluating large language models (LLMs) for clinical decision support requires capturing complex reasoning that traditional automated metrics often miss. In this study, we evaluate LLaMA-3.3-70B and Gemma-4-31B on the hypotheses of clinical diagnosis and treatment recommendation using 2,000 patient records from the MIMIC-IV database. We assess performance using lexical overlap (BLEU/ROUGE), prescription drug alignment, temperature sensitivity, and local interpretability via LIME. To capture qualitative clinical value, an autonomous evaluator (GPT-OSS 120B) ranked the output according to accuracy, completeness, safety, and utility. Gemma-4-31B secured a head-to-head win rate of 97.9% and a median clinical quality score of 85% compared to LLaMA's 52%. Crucially, Gemma achieved these results despite scoring lower on BLEU (0.015 vs. 0.030) and matching LLaMA on ROUGE-1. This divergence demonstrates that the precision metrics n -gram heavily penalize clinical verbosity and detailed justifications, which proves inadequate for the generation of open-ended medical texts.

1 Introduction

Incorporating artificial intelligence into clinical workflows remains a primary goal of biomedical informatics. Electronic health records (EHRs) track complete patient histories, including diagnoses, laboratory findings, vitals, and treatments. Although interpreting these records typically required years of specialized medical training, modern large language models (LLMs) trained on diverse biomedical datasets show significant potential in translating these raw clinical data into actionable treatment insights.

Although LLMs have excelled on structured medical benchmarks, such as passing the USMLE examination [13], multiple-choice testing does not reflect the true complexity of live clinical environments. Real-world medical practice demands open-ended reasoning: formulating accurate differential diagnoses, outlining treatment strategies, mitigating drug interactions, and highlighting patient safety concerns.

The recent availability of capable open-weight models, including Meta's LLaMA [3] and Google's Gemma [4] series, makes it possible to deploy inference pipelines locally within academic and clinical constraints. However, comparative evaluations of these models in realistic un-curated health data remain limited [12, 18].

To evaluate these models under realistic conditions, we developed an evaluation pipeline using real-world records from the MIMIC-IV clinical database [9]. The core contributions of this study are:

- **Pipeline Architecture:** We designed a scalable data ingestion system that queries a local DuckDB-indexed instance of the MIMIC-IV dataset, structuring dense EHR inputs into clinical prompts for 2,000 patient records.
- **Comprehensive Benchmarking:** We evaluated model responses using classical text similarity metrics (BLEU and ROUGE-1/2/L), drug prescription overlap ratios, and a multi-dimensional assessment (covering accuracy, completeness, safety, and utility) managed by an automated GPT-OSS 120B judge [19].
- **Sensitivity and Interpretability Analysis:** We evaluated the impact of model generation temperature on Gemma-4-31B performance using a subset of 1,000 patients and applied LIME framework analysis to map how specific prompt segments steer model output.

2 Background and Related Work

2.1 NLP in Clinical Medicine and Electronic Health Records

The application of natural language processing to clinical text began with basic rule-based systems designed to catch named entities in discharge summaries. The introduction of transformer architectures fundamentally transformed the field. Models like BioBERT [10] and ClinicalBERT [1] demonstrated that pretraining language models directly on biomedical and clinical text drastically improves performance on downstream tasks such as relation extraction, question answering, and named entity recognition (NER).

The digitization of healthcare has generated many large Electronic Health Records (EHR) repositories, which serve as crucial training and evaluation data for clinical machine learning systems. Among these, the Medical Information Mart for Intensive Care (MIMIC) database has become a popular reference, as it is a comprehensive and freely available repository containing de-identified data for tens of thousands of admissions to intensive care units (ICU) [8].

In the past, researchers using the MIMIC database have focused mainly on training machine learning models for specific single tasks, such as forecasting the length of stay in a patient's intensive care unit [18]. Even if these older deep learning models were useful for such tasks, they were limited because they could only do one job at a time. They were unable to examine the entire medical profile of a patient to connect the dots and provide a full clinical recommendation.

2.2 LLMs on Medical Tasks

Recent advances have shifted the landscape toward generative modeling. Instead of training niche classification networks, researchers increasingly use general-purpose frontier models for clinical diagnosis and treatment planning [12]. Although closed-source models initially dominated early clinical feasibility studies, privacy constraints and regulations relating to data sovereignty in the health-care sector have led to growing interest in open-source alternatives.

Open-weight models, such as Google’s Gemma architecture, offer the possibility of deploying these models locally, ensuring strict compliance with patient data privacy. Specific architectural optimizations and alignment strategies significantly impact the clinical utility of a model, particularly when synthesizing highly technical structures containing laboratory values [12].

Singhal et al. [5] show that instruction-tuned language models encode substantial clinical knowledge and achieve competitive performance on medical question-answering. Nori et al. [13] demonstrate GPT-4’s ability to pass standardized medical licensing examinations, while Jin et al. [7] present large-scale question-answering datasets demonstrating the gap between model capabilities and expert human performance.

However, the performance of free-form clinical generation situated within real patient records remains considerably less studied. Our work directly targets this gap by constructing prompts from structured EHR data rather than from curated examination questions.

2.3 Automated Evaluation and Lexical Metrics

The evaluation of free-form clinical text generated by LLMs remains a contentious challenge. Traditionally, machine translation and summarization tasks have relied on automated lexical overlap metrics, primarily Bilingual Evaluation Understudy (BLEU) [15] and Recall-Oriented Understudy for Gisting Evaluation (ROUGE) [11]. These metrics quantify the precise n -gram overlap between a generated output and a reference string.

However, recent literature increasingly highlights a critical limitation of these metrics: their inefficiency in replicating human judgment of results [14]. A model could generate an accurately detailed and safer treatment plan that diverges from the bare style of a reference text, resulting in artificially depressed BLEU and ROUGE scores. Alternatively, a superficial lexical match may result in shorter, less descriptive results simply because they contain fewer tokens that are likely to be penalized by BLEU and ROUGE metrics.

To overcome this problem and capture nuance, contemporary studies utilize the “LLM-as-a-judge” framework [20]. In these setups, an independent model evaluates the generated text according to standards such as helpfulness, accuracy, relevance, and completeness [20] offering an evaluation pipeline that correlates more closely with human expert judgment than traditional string-matching metrics.

The authors of these studies demonstrate strong agreement between GPT-4 judgments and human annotations on open-ended tasks. This is why we chose to use GPT-OSS 120B as our evaluator, scoring the output of the model on four dimensions of clinical quality in all 2,000 patients.

2.4 Explainable AI (XAI) in Clinical Decision Support

Because Large Language Models function essentially as black boxes, deploying them in high-stakes clinical decision support systems introduces profound ethical and operational risks. Medical professionals cannot safely rely on AI recommendations without understanding the underlying reasoning or knowing which parameters of the patient drove a specific conclusion. This is why explainable AI (XAI) becomes an essential element.

Local Interpretable Model-agnostic Explanations (LIME) is a framework designed to address this lack of transparency [17]. LIME operates by perturbing the input features of a given instance, such as masking specific laboratory values or ICD codes, and observing how these changes alter the model’s output distribution. By training an interpretable local surrogate model for these perturbed features, LIME quantifies the importance of individual inputs. Integrating XAI into clinical LLM evaluation pipelines ensures that models can be audited for clinical soundness, verifying that they are making decisions based on genuine medical indicators rather than superficial variables or demographic bias.

3 Methodology and Design

3.1 Data Preparation

All clinical data utilized in this study are derived from the MIMIC-IV v3.1 database [8], a freely accessible collection of de-identified health records of over 40,000 patients admitted to critical care units, locally indexed using DuckDB.

Patient eligibility requires at least one ICD diagnosis code, at least one laboratory event, and at least one prescription record. From the eligible pool, a reproducible random cohort of 2,000 patients is drawn using a seeded shuffle (seed = 42). Cohort identifiers are persisted in CSV to guaranty identical patient selection across pipeline restarts.

3.2 Two-Phase Data Loading

To handle massive amounts of data under strict memory constraints, we designed a two-phase database loading layer leveraging a local DuckDB configuration.

- (1) **Phase 1:** In the beginning, a lightweight SQL query isolates eligible hospital admission identifiers (`hadm_id`). To ensure the reproducibility of the calculations, a random sampling step shuffles the cohort using a set seed (`RANDOM_SEED = 42`) and extracts a standardized set of patients, stored locally in the file `cohort_ids.csv`. This acts as a fixed reference, preventing data loss or cohort drift during interrupted executions
- (2) **Phase 2:** To maintain stable system RAM usage during high-throughput execution, data processing is divided into sequential blocks (`BATCH_SIZE = 50`). The pipeline dynamically retrieves records (including longitudinal lab events, vitals, and medications) for the active slice, freeing the associated data frames from memory once the batch processing step is complete.

3.3 Prompt Construction

For each patient, the context profile is compiled into a unified textual prompt to simulate an electronic health record summary. The system architecture enforces a *Senior Attending Physician Persona*. The structure of the injected context contains 8 sections as follows:

- **Demographics:** Baseline parameters of the patient, such as biological sex and anchor age.
- **History of Diagnoses:** Chronologically ordered historical conditions matching the standard International Classification of Diseases (ICD) nomenclature.
- **Lab Results and ICU vitals:** The 15 most recent numeric lab events (such as blood counts, for example) paired with up to 12 vital signs from the intensive care unit (ICU) drawn directly from continuous observation charts (chartevents).
- **Clinical Procedures and Microbiology:** Chronological logs of surgical or medical procedures and culture outcomes.
- **Duration of Stay in the ICU**

The task instruction requires each model to respond in exactly four structured parts like this:

- (1) **Diagnosis Hypotheses:** List 3 to 5 plausible differential or co-morbid diagnoses and support each one with a one-sentence clinical rationale.
- (2) **Recommended Treatments:** Provide specific drug names (and typical doses, where reasonable), also include non-pharmacological recommendations if relevant, and justify briefly.
- (3) **Safety Flags:** Identify any concerns about interactions, contraindications, or monitoring needs.
- (4) **Confidence Level:** Give your confidence level (Low | Medium | High) with a one-sentence explanation.

To balance structural determinism with medical reasoning, the models generate outputs under a moderate generation temperature ($TEMPERATURE = 0.3$), which allows the outputs to be slightly more creative, and a set limit for the response length ($MAX_TOKENS = 2500$).

3.4 Models

LLaMA-3.3-70B-Versatile is Meta’s 70-billion-parameter decoder-only transformer [3], accessible via the Groq inference API. **Gemma-4-31B-IT** is Google’s 31-billion-parameter instruction-tuned model [4], accessed via the Google Generative AI API. Both models use a temperature of $T = 0.3$ and a maximum of 2,500 output tokens. The autonomous evaluator **GPT-OSS 120B** is queried at temperature $T = 0.1$ with up to 3,000 tokens.

3.5 Drug Overlap Analysis

The true prescribed medications are retrieved from the MIMIC-IV prescription table. A fuzzy string matching algorithm (Python `difflib SequenceMatcher`, threshold ≥ 0.75) compares drug tokens extracted from model outputs with the ground-truth prescription list, yielding absolute match counts and recall ratios normalized by the number of true prescriptions.

3.6 Temperature Sensitivity Experiment

Gemma-4-31B is re-queried in a subset of 1,000 patients at three temperature settings

$$T \in 0.1, 0.5, 1.0.$$

BLEU and ROUGE-1 scores are calculated against patient-concatenated ICD diagnosis descriptions, producing 3,000 evaluation pairs.

4 Evaluation Strategy

4.1 Lexical Similarity

The pipeline calculates BLEU scores using NLTK’s smoothing function alongside ROUGE-1, ROUGE-2, and ROUGE-L [2] scores using a standard porter stemmer.

4.2 LLM-as-a-Judge

To capture medical precision that lexical string matching overlooks, we use the `openai/gpt-oss-120b` high-capacity model that serves as an automated clinical judge. The judge analyzes both outputs, assigning scores according to four critical axes:

- (1) **Clinical Accuracy:** Correctness of differential diagnoses, recommended drugs, and dosing.
- (2) **Completeness:** Coverage of all clinically relevant aspects without omitting critical information.
- (3) **Safety:** Identification of drug interactions, contraindications, and safety warnings.
- (4) **Practical Usefulness:** Clarity, structure, and applicability for a clinical practitioner.

Each axis has a ceiling score of 25, resulting in a total maximum score of 100. To evaluate accuracy, the system compares its top drug recommendations with real clinical prescriptions. It uses fuzzy matching with a 75% similarity threshold ($FUZZY_THRESHOLD = 0.75$) to account for minor naming variations when calculating Precision@K [16].

The evaluator additionally designates a head-to-head winner (model1, model2, or tie) and provides a short comparison rationale returned as structured JSON parsed pro-grammatically.

4.3 LIME Usage

To verify the trustworthiness of the main reasoning pipelines, we implemented an explainability module using LIME. Standard LIME on text requires invoking the base model thousands of times per sample, which introduces unsustainable computational and financial API bottlenecks.

To avoid this, we used the TF-IDF cosine similarity as a proxy scoring function to score perturbations. Let Y_{orig} represent the original, unperturbed text output generated by one of the models for a given prompt. When LIME perturbs a clinical prompt X to generate a subset prompt X_{pert} , a local vectorizer converts the text into a sublinear term frequency-inverse document frequency space (TF-IDF) [6].

The proxy scoring function computes the cosine similarity between the perturbed text space and the static reference vector of

Y_{orig} :

$$\text{Similarity}(X_{pert}, Y_{orig}) = \frac{\vec{V}(X_{pert}) \cdot \vec{V}(Y_{orig})}{\|\vec{V}(X_{pert})\| \|\vec{V}(Y_{orig})\|} \quad (1)$$

Tokens whose masking drastically drops the cosine similarity to the original output are assigned positive LIME weights, pinpointing them as influential drivers of the model’s conclusion. Finally, individual token weights are aggregated using regular expressions in prompt headers (PROMPT_SECTIONS) to build a layout of importance for our localized clinical features.

5 Results and Analysis

Figure 1 presents the full evaluation dashboard. Table 1 summarizes quantitative results.

Table 1: Comparative results ($n = 2,000$). BLEU and ROUGE values are medians. Evaluator scores are medians as percentages of maximum.

Metric	LLaMA-70B	Gemma-31B
Output length (chars)	2,200	5,000
Drug overlap (count)	1	2
Drug overlap (% true Rx)	5%	12%
BLEU	0.030	0.015
ROUGE-1	0.210	0.210
ROUGE-2	0.070	0.050
ROUGE-L	0.140	0.120
Evaluator total (%)	52	85
Accuracy (%)	60	85
Safety (%)	52	90
Completeness (%)	47	80
Usefulness (%)	52	85
Head-to-head wins	38 (1.9%)	1,959 (97.9%)

5.1 Output Verbosity

A pronounced difference in output verbosity characterizes the two models. LLaMA-3.3-70B produces consistently concise outputs (median $\approx 2,200$ characters, IQR 1,800-2,300), while Gemma-4-31B generates substantially longer responses (median $\approx 5,000$ characters, IQR 3,500-5,000). This difference in output length is the primary driver of the divergence in n-gram similarity scores discussed below.

5.2 Drug Overlap

Both models achieve modest drug overlap counts (median 1 matched drug for LLaMA, 2 for Gemma), reflecting the inherent difficulty of exact pharmacological recommendation from ICD diagnosis codes alone. The percentage of drug overlap is approximately 5% for LLaMA and 12% for Gemma, consistent with the findings in related studies of pharmacology of LLM. Gemma’s wider output distribution and higher count suggest that its more comprehensive outputs successfully recover additional relevant medications.

5.3 Lexical Similarity and the Brevity Effect

LLaMA-3.3-70B achieves a higher median BLEU (0.030 vs. 0.015) than Gemma-4-31B, while the ROUGE-1 scores are nearly equal (≈ 0.210). The explanation lies in the *brevity effect*: LLaMA’s shorter, diagnosis-focused outputs contain a higher proportion of ICD-related vocabulary, yielding greater precision overlap with the concise ICD reference strings. Gemma’s comprehensive outputs, which include treatment plans, drug interaction warnings, and procedural guidance, contain proportionally less diagnosis-specific vocabulary and are penalized by n-gram precision metrics like BLEU.

This result has a direct methodological implication: BLEU and ROUGE should not serve as primary quality indicators for open-ended clinical generation tasks where reference strings are concise ICD descriptions.

5.4 Clinical Quality Evaluation

The GPT-OSS 120B evaluator reveals a decisive advantage for Gemma-4-31B in all four quality dimensions. Gemma achieves a median total score of approximately 85% versus LLaMA’s 52%. In head-to-head comparisons, Gemma wins 1,959 out of 2,000 evaluations (97.9%), LLaMA wins 38 (1.9%) and 3 result in a tie.

The greatest performance gaps appear in Safety (Gemma 90% vs. LLaMA 52%) and Usefulness (Gemma 85% vs. LLaMA 52%), confirming that Gemma’s verbosity translates into genuine clinical value and safer recommendations. Completeness (Gemma 80% vs. LLaMA 47%) and Accuracy (Gemma 85% vs. LLaMA 60%) also show substantial differences.

5.5 Temperature Sensitivity

Table 2 shows the effect of temperature on Gemma output quality. BLEU scores remain uniformly low (≈ 0.007) at all three temperatures. ROUGE-1 also remains consistently stable (≈ 0.110) suggesting that the generation quality measured by lexical metrics is relatively insensitive to temperature. Overall, Gemma-4-31B is largely robust to this hyperparameter within the tested range.

Table 2: Temperature sensitivity for Gemma-4-31B ($n = 1,000$ per setting). Values are medians.

Temperature	BLEU	ROUGE-1
0.1	0.007	0.110
0.5	0.007	0.110
1.0	0.007	0.110

6 LIME Explainability

Figure 2 displays the most influential words globally aggregated in all sampled patients for each model. For both LLaMA-3.3-70B and Gemma-4-31B, tokens from the Diagnoses and Laboratory Results sections consistently rank among the most influential. This confirms that both models are anchored primarily on a clinical diagnostic context, rather than on demographic information or procedural history. Section-level heatmap analysis reveals a notable difference: LLaMA’s outputs are most strongly influenced by primary diagnosis terminology, reflecting its concise, diagnosis-centric generation style. Gemma exhibits more distributed influence

across the Diagnoses, Laboratory Results, and Vital Signs sections, consistent with its broader, more integrative output behavior.

7 Discussion and Limitations

7.1 Discussion and Limitations

Three key findings emerge from this study. First, the brevity effect demonstrates that standard automatic metrics can produce misleading model rankings for clinical text generation. Shorter outputs that closely mirror reference diagnosis strings score higher on BLEU and ROUGE, despite being clinically inferior. This result advocates for multidimensional LLM-based evaluation as the standard for clinical text generation benchmarks. Second, the autonomous clinical evaluator strongly prefers the comprehensive output style of Gemma-4-31B, characterized by detailed treatment justifications, explicit safety warnings, and structured confidence ratings by all four quality dimensions. The 97.9% head-to-head win rate across 2,000 patients is a robust result that substantially reduces the influence of prompt-level variance. Third, the temperature sensitivity experiment shows that Gemma-4-31B consistently produces quality outputs across

$$T \in 0.1, 0.5, 1.0.$$

, with only marginal variation in the lexical diversity metrics. This robustness is practically significant for clinical deployment pipelines operating in fixed inference configurations.

The performance paradox observed between lexical overlap scoring and LLM alignment requires a deeper analysis of architectural constraints. To validate that the models engaged in genuine clinical reasoning rather than exploiting superficial stylistic shortcuts, we looked to the importance of the features computed by our localized LIME pipeline.

As illustrated in Figure 2, the token weight distribution in the prompt sections confirms a high degree of clinical alignment. For both LLaMA-3.3-70B and Gemma-4-31B, the LIME engine assigned the highest importance weights to the characteristics of the tokens residing within the subsections LAB_EVENTS and VITAL_SIGNS. Crucially, administrative text segments exhibited a negligible impact on the output similarity space. This pattern confirms that the underlying reasoning engines focus on clinical data parameters to formulate their diagnostic hypotheses and treatment suggestions, providing a strong baseline for transparency and accountability in clinical deployment.

7.2 The Validity and Implications of AI-by-AI Evaluation

A central point of this study is the deployment of a high-capacity model (gpt-oss-120b) that acts as an automated clinical judge to score free-form generative output. We know that using an LLM-as-a-judge offers a scalable, reproducible alternative to costly human clinician reviews, but it introduces important methodological challenges.

An inherent limitation of this system is the risk of what is called “self-reinforcement” bias or systemic blind spots. An AI judge might favor smooth linguistic structure and persuasive clinical prose, potentially misinterpreting an eloquently phrased but clinically incorrect hypothesis as a high-quality recommendation. To address

this vulnerability, our methodology pairs the qualitative LLM judge with a deterministic drug-verification module using fuzzy string matching. By checking generated prescriptions against authentic hospital pharmacy records at a strict similarity threshold of 0.75, we anchored the open-ended classification system to empirical clinical data, successfully balancing flexible AI judgment with rigid factual grounding.

7.3 Hallucination Containment

In high-stakes ICU environments, text generation errors present an immediate clinical risk. Hallucinations generally manifest in two ways: intrinsic hallucinations, where a model misinterprets input variables (in our case it could be misreading a laboratory result), and extrinsic hallucinations, where a model introduces clinical interventions or diagnoses completely detached from the patient’s record.

Our pipeline minimized these errors by combining strict prompt constraints with narrow hyperparameter boundaries. Setting the generation temperature at 0.3 restricted token selection to a low-entropy distribution, preserving enough clinical flexibility to correlate complex variables, such as linking elevated white blood cell counts and positive blood cultures to a sepsis hypothesis. In addition, this setting prevented stylistic deterioration and unsupported factual fabrications observed at higher temperatures. Furthermore, requiring a single-sentence clinical justification for each additional diagnosis acted as a foundation mechanism, making it easy to audit the results against the source EHR files.

7.4 Limitations

The LIME explainability analysis employs a lexical overlap proxy rather than true semantic attribution. Future work should apply gradient-based attribution methods for more rigorous interpretability. Drug overlap relies on fuzzy string matching, which may both over-count matches between pharmacologically unrelated drugs sharing similar character sequences and undercount paraphrased or abbreviated drug names. A standardized pharmacological ontology such as RxNorm would yield more precise recall measurements. The autonomous evaluator can introduce preference biases, potentially favoring well-structured, verbose output independent of actual clinical correctness. The validation of experts in humans by practicing clinicians would substantially strengthen the validity of the evaluation. Finally, our cohort represents a single institution and a historical time period (2008-2024). Generalization to contemporary clinical settings and diverse healthcare systems requires validation on more recent and multi-institutional datasets.

7.5 Additional concerns

The project’s evaluation framework highlights a critical tension between traditional automated metrics and the clinical utility of large language models (LLMs). Lexical metrics such as BLEU and ROUGE are insufficient for the medical domain; by measuring strict n -gram overlap against standard references, they penalize model verbosity. However, in clinical contexts, this additional text often contains critical substance and safety margins, causing models like Gemma-4-31B to be unfairly penalized despite their real-world utility.

This gap underscores the need to transition from syntactic metrics to autonomous approaches that evaluate qualitative dimensions. Furthermore, explainability tools such as LIME offer valuable global insights, but have distinct limitations. LIME acts as a lexical proxy, mapping token sensitivity rather than exposing underlying reasoning, hallucinations, or procedural errors. Ultimately, relying on BLEU, ROUGE, and LIME creates a disconnected collection of validation tools rather than a cohesive, hypothesis-driven framework.

Future work must align evaluation metrics with specific clinical objectives, ensuring that frameworks assess diagnostic integrity rather than superficial textual overlap.

8 Conclusion and Future Work

This paper presents Project Omega, a comprehensive evaluation of LLaMA-3.3-70B and Gemma-4-31B to hypothesize a clinical diagnosis and recommend treatment in 2,000 admissions from MIMIC-IV patients. Our pipeline combines drug overlap analysis, lexical similarity metrics, LLM-based clinical quality assessment, temperature sensitivity experiments, and LIME explainability analysis into a reproducible evaluation framework. Gemma-4-31B substantially outperforms LLaMA-3.3-70B in clinical quality, winning 97.9% of head-to-head evaluations, while simultaneously scoring lower on the BLEU and ROUGE metrics. This contradiction is resolved when output verbosity is considered: Gemma’s longer, more comprehensive outputs are penalized by n-gram metrics that reward lexical conciseness relative to short ICD reference strings. This finding has direct implications for how clinical text generation systems should be evaluated and motivates the adoption of multi-dimensional LLM-based evaluation frameworks alongside traditional metrics.

We successfully established and evaluated a local clinical decision support pipeline that uses open-weight Large Language Models (LLMs) on real-world EHR profiles from the MIMIC-IV v3.1 database. By intentionally designing an indexed batch-streamed database infrastructure, the framework handled tens of thousands of complex ICU data strings while maintaining a completely flat hardware memory footprint. Furthermore, the validation of our optimized, API-free local LIME proxy interpreter proved that both underlying language models relied heavily on relevant clinical parameters (vitals and lab events) rather than administrative tokens or superficial prompt fields to make their conclusions.

There are multiple compelling pathways for future work following this project:

- **Loop in human clinicians:** Integrating professional medical clinicians to perform a blind review of LLM outputs to rigorously cross-verify and map the precise alignment correlation between the AI judge and human experts.
- **Vector extensions with images** Expanding the ingestion pipeline beyond structured alphanumeric tabular streams to integrate vector embeddings of raw medical imaging transcripts (radiology reports).
- **Evaluation of contemporary multi-institutional datasets:** Testing generalization across diverse healthcare systems and more recent data collections.
- **Targeted evaluation of safety-critical subpopulations:** Focused evaluation of pediatric, oncological, and other high-risk patient groups.

- **Fine-tuning:** Domain-adaptive training to improve diagnostic accuracy and reduce hallucination rates.

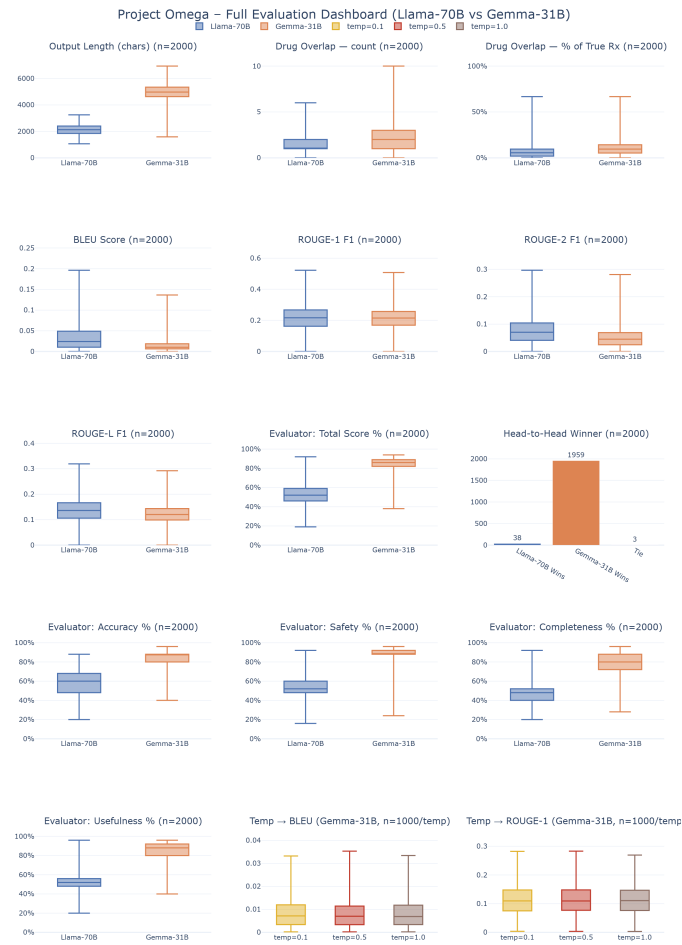


Figure 1: Full evaluation dashboard for 2,000 MIMIC-IV patients. Rows 1–2: output statistics and lexical similarity metrics. Row 3: head-to-head winner counts and evaluator total scores. Rows 4–5: evaluator subscores and temperature sensitivity analysis (Gemma-4-31B, $n = 1,000$ per temperature).

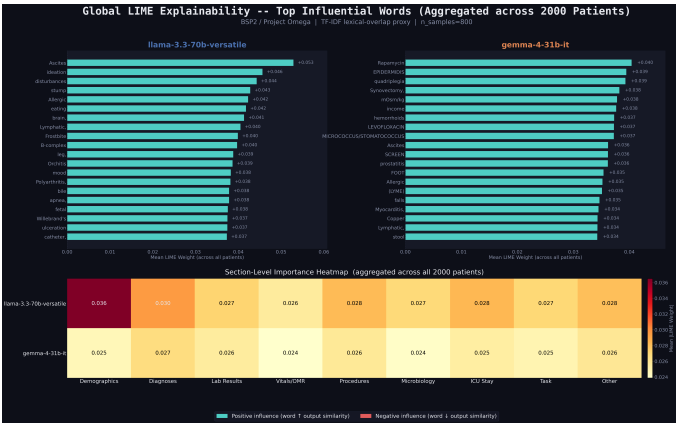


Figure 2: Global LIME explainability results for LLaMA-3.3-70B (left) and Gemma-4-31B (right), aggregated across 50 patients each. Bars show mean absolute influence weight; color indicates direction (teal: positive, coral: negative). Bottom panels show section-level importance heatmaps.

9 Work split

Nina Vojtaššák:

- **Exploratory Research:** Conducted initial research on MIMIC-IV dataset functions and architecture, drafting a brief used to align the team’s technical scope.
- **Review of Academic Literature:** Researched and analyzed similar medical LLM studies to establish the comparative framework for the final report. Relevant references to academic literature were used where necessary.
- **Project Infrastructure:** Initialized the team’s GitHub repository and organized the foundational documentation to establish a collaborative workflow.
- **Script Prototyping:** Developed the early-stage Python script “preprocess_mimic.py” to clean and parse the MIMIC-IV demo dataset, allowing the team to begin preliminary testing while awaiting access to the full dataset.
- **Algorithmic Explorations (Unused Features):** Programmed experimental features, including an image processing module (“images _mimic_v1.py” to include images in the table of relevant data, “pipelines-images-brainstorm.py” to add image processing to the pipeline), and a patient readmission function “readmissions.py”, which served as valuable early proof-of-concepts despite being scoped out of the final model due to time constraints.
- **Code Review and Support:** Acted as a peer-reviewer by auditing team-written code to ensure logical correctness. Served as a technical resource within the group, explaining important code segments to keep all team members aligned.
- **Writing:** Wrote the following sections of the report: Section 2, Section 3, Section 4, Section 5, Section 7, Section 8. Contributed to the writing of the presentation.
- **Formatting:** Properly formatted the final report.

Can Yildiz:

- **Designed Pipeline:** Built an automated system for API requests to ensure uninterrupted processing of 4,000 patients.
- **Integrated Model 1 and Model 1:** Llama 3.3 70B via multiple API keys, and Gemma 4 31B via Google Gemini API.
- **Integrated Evaluator Model:** GPT-OSS 120B via multiple API keys.
- **Implemented Error Handling:** Added robust catch mechanisms for API limits (429), exhausted credits (402), missing models (404) and permission denials (403), triggering automatic provider switching.
- **Updated Hyperparameter Experiments:** Wrote “temperature.py” to use Gemma 4 31B and adjusted the testing temperatures to [0.1, 0.5, 1.0].
- **Aligned Analysis Scripts:** Wrote “bleu_rouge_metrics.py”, “graph_maker.py”, and “lime_analyzer.py” to reflect the new pipeline architecture, model names, and chart labels.
- **Added Resume Functionality:** Implemented logic to track processed patients and seamlessly resume execution after interruptions.
- **Executed the Pipeline and Data Collection:** Ran the end to end patient analysis pipeline, effectively processing 2,000 cases to generate the final LLM response and evaluation datasets.
- **Data Extraction:** Successfully ran all analytical scripts to extract final performance metrics, generation scores, and explainability (LIME) results for the study.
- **Version Control and Deployment:** Uploaded and managed the code base and project files on GitHub for version control and collaborative access.
- **Data Preprocessing and Local Database Setup:** Developed “init_db.py” to parse raw MIMIC-IV CSV files and initialize a high-performance local DuckDB database, enabling fast and reproducible extraction of the cohort of 2,000 patients.
- **Prompt Engineering and EHR Data Structuring:** Designed the structured prompt templates that dynamically map various “Electronic Health Records” (including ICD diagnoses, laboratory results, ICU vitals, and procedures) into cohesive, standardized inputs for the LLMs utilized.
- **Writing:** Wrote the following sections of the report: Section 1, Section 5, Section, 6, Section 8 and Abstract.

Tuna Karakus:

- **Database:** Studied the MIMIC-IV dataset and understood its tables and relationships.
- **Code Review:** Reviewed the implementations in the code-base and identified issues.
- **Academic Literature and Methodology:** Read the relevant literature to understand the motivation of the evaluation methods used in the project.
- **Presentation and Review:** Contributed to the preparation of the presentation and reviewed the report for technical consistency.

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